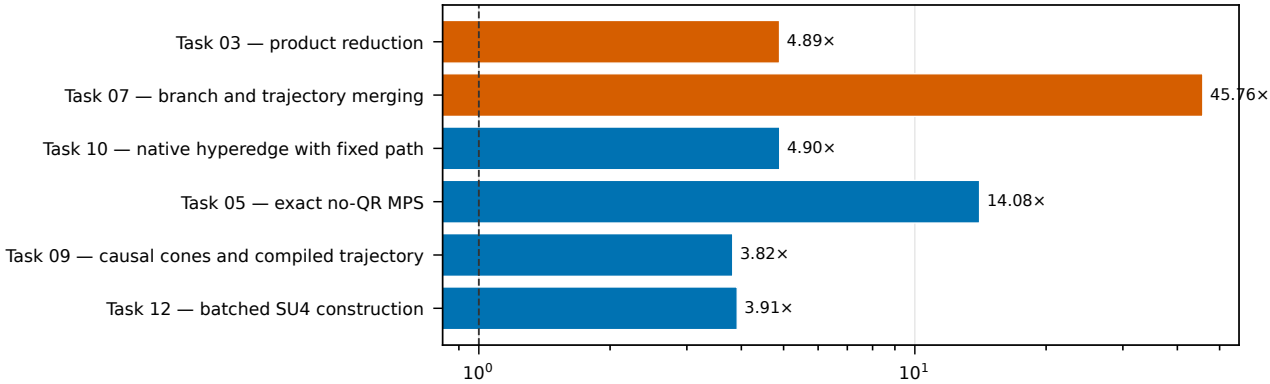


Take-home optimization insights



Reported end-to-end speedup (x; log scale)

Orange: exact challenge-design reductions (Tasks 03 and 07); blue: framework-native implementation improvements. Factor rows are direct ablations and are not additive